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KEZORAL
Tablet 200mg
Ketoconazole 200mg

DESCRIPTION

A 10mm diameter, round, flat, scored, marked "K 200" white tablet.

INDICATIONS

Kezoral (ketoconazole) Tablets should be used only when other effective antigfungal therapy is not available or tolerated and the potential benefits are considered to outweigh the potential risks.
Kezoral (ketoconazole) Tablets are indicated for treatment of the systemic fungal infections in patients who have failed or who are intolerant to other therapies:
- Blastomycosis
- Coccidioidomycosis
- Histoplasmosis
- Chromomycosis
- Paracoccidioidomycosis

Kezoral (ketoconazole) Tablets should not be used for fungal meningitis because it penetrates poorly into the cerebrospinal fluid.

PHARMACODYNAMICS

Ketoconazole is fungistatic, and may be fungicidal, depending on concentration. Ketoconazole interferes with cytochrome P-450 activity, which is necessary for the demethylation of 14-alpha-methysterols to ergosterol. Ergosterol, the principal sterol in the fungal cell membrane, becomes depleted. This damages the cell membrane,producing alterations in membrane functions and permeability. In Candida albicans, ketoconazole inhibits transfomation of blastospores into invasive mycelial form.

PHARMACOKINETICS

The absorption of ketoconazole from the gastrointestinal tract is variable and increases with decreasing stomach pH. Peak plasma concentrations of up to 7 mcg per ml have been obtained two hours after administration of 200 mg by mouth. It is extensively bound to plasma proteins. Penetration into cerebrospinal fluid is poor following oral administration. The elimination of ketoconazole is reported to be biphasic, with a terminal half-life of about 8 hours. Ketoconazole is extensively metabolised in the liver to inactive metabolites. It is excreted as metabolites and unchanged drug chiefly in the faeces; some is excreted in the urine. Ketoconazole has been shown to suppress corticosteroid secretion and lower serum testosterone concentrations.

DOSAGE AND ADMINISTRATION

The usual dose for treatment and prophylaxis of fungal infections is 200 mg once daily taken with food. This may be increased to 400 mg if an adequate response is not obtained. Children may be given approximately 3 mg per kg body weight daily, or for children aged 5 to 12 years, 100 mg daily. The duration of treatment has not been established for most fungal infections, but should be continued until at least one week after all symptoms have disappeared and all cultures turn out negative some infections may require at least 6 months of therapy.
The type of organism responsible for the infection should be identified; however therapy may be initiated prior to obtaining these results, when clinically warranted.
Ketoconazole tablets should be taken during meals for maximal absorption.

Ketoconazole tablets are indicated for patients who have failed or who are intolerant to other therapies. (see Indications)

OVERDOSAGE

In the event of accidental overdosage, supportive measures including gastric lavage with sodium bicarbonate, should be employed.
Up-to-date information on the treatment of overdose can be obtained from The National Poison Centre, Universiti Sains Malaysia.

TOXICOLOGY

Paediatrics : Several cases of hepatitis have been reported in children who have taken ketoconazole. Appropriate studies on the relationship of age to the effects of ketoconazole have not been performed in children up to 2 years of age. However, no paediatrics-specific problems have been documented to date in children over 2 years of age.
Geriatrics : No information is available on the relationship of age to the effects of this medicine in geriatric patients.

Carcinogenicity/Tumorigenicity : Long-term feeding studies in Swiss albino mice and in Wistar rats have not shown evidence of oncogenesis.
Mutagenicity : Dominant lethal mutation tests have not shown mutation in any stage of germ cell development in male and female mice given single oral doses of ketoconazole as high as 80 mg/kg. In addition, the Ames/Salmonella microsomal activator assays have not shown evidence of mutagenicity.

Pregnancy/Reproduction
Fertility : Ketoconazole has been shown to decrease or abolish serum testosterone concentration when used in high doses (eg. 800 mg to 1.6 g daily). Ketoconazole has also been shown to cause menstrual irregularities, oligospermia, azoospermia, impotence, and decreased male libido.

Pregnancy : Ketoconazole crosses the placenta. Adequate and well-controlled studies in humans have not been done. Studies in rats given doses of 80 mg/kg/day (10 times the maximum recommended human dose) have shown ketoconazole to be teratogenic, causing syndactyly and oligodactyly. Ketoconazole has also been shown to be embryotoxic in rats given doses greater than 80 mg/kg during the first trimester. ***Ketoconazole has also been shown to cause dystocia in rats given doses greater than 10 mg/kg (greater than 1.25 times the maximum recommended human dose [MRHD] during the third trimester. (FDA Pregnancy Category C)***

Breast-feeding : Ketoconazole when taken orally is distributed into the breast milk. However, it is not known whether ketoconazole, applied topically on a regular basis, is absorbed systemically in sufficient amounts to be distributed into breast milk in detectable quantities. However, no systemic absorption was detected following a single application to the chest, back and arms of healthy volunteers. Therefore, topical ketoconazole is unlikely to be distributed into breast milk in significant amounts to cause adverse effects in the nursing infant.

CONTRAINDICATIONS

- In patients with acute or chronic liver disease.
- Hypersensitivity to the drug or excipient.
- Use in pregnancy only if the potential advantage justifies the possible risk to the foetus.

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WARNINGS & PRECAUTIONS

Because of the risk for serious hepatotoxicity, (Kezoral Tablets) should be used only when the potential benefits are considered to outweigh the potential risks, taking into consideration the availability of other effective antifungal therapy.
Assess liver function, prior to treatment to rule out acute or chronic liver disease, and monitor at frequent and regular intervals during treatment, and at the first signs or symptoms of possible hepatotoxicity.

Hepatotoxicity

Very rare cases of serious hepatotoxicity, including cases with fatal outcome or requiring liver transplantation have occurred with the use of oral ketoconazole. Some patients had no obvious risk factors for liver disease. Cases have been reported that occurred within the first month of treatment, including some within the first week.
The cumulative dose of the treatment is a risk factor for serious hepatotoxicity. Factors which may increase the risk of hepatitis are prolonged treatment with ketoconazole tablets, females over 50 years of age, previous treatment with griseofulvin, a history of liver disease, known drug intolerance and concurrent use of medication which compromises liver function. A period of one month should be allowed between cessation of griseofulvin treatment and commencement treatment with ketoconazole tablets because of an apparent association between recent griseofulvin therapy and hepatic reactions to ketoconazole tablets.
Monitor liver function in all patients receiving treatment with ketoconazole tablets (see Monitoring of hepatic function)
Patients should be instructed to promptly report to their physician signs and symptoms suggestive of hepatitis such as anorexia, nausea, vomiting, fatigue, jaundice, abdominal pain or dark urine. In these patients, treatment should be stopped immediately and liver function should be conducted.
Monitoring of hepatic function
Monitoring liver function in all patients receiving treatment with ketoconazole tablets. Monitor liver function prior to treatment to rule out acute or chronic liver disease (see CONTRAINDICATIONS), after two weeks of treatment and then on a monthly basis and at the first signs or symptoms of possible hepatic toxicity. When the liver function tests indicate liver injury, the treatment should be stopped immeadiately.
A risk and benefit evaluation should be made before oral ketoconazole is used in cases of non-life threatening diseases requiring long treatment periods.
In patients with elevated liver enzymes, or who have experienced liver toxicity with other drugs, treatment should not be started unless the expected benefit exceeds the risk of hepatic injury. In such cases, close monitoring of the liver enzymes is necessary.

SIDE EFFECTS

Kezoral is well tolerated. In rare cases, anaphylaxis has been reported after the first dose. Several cases of hypersensitivity reactions including urticaria have also been reported. However, the most frequent adverse reactions were nausea and/or vomiting, abdominal pain and pruritus. Symptoms of headache, dizziness, somnolence, fever and chills, photophobia, diarrhoea, gynaecomastia have been reported in investigational studies with the drug at dosage above those currently approved.
Post-marketing experience
Hepato-billiary disorders
Very rare: serious hepatotoxicity, including hepatitis cholestatic, biopsy-confirmed hepatic necrosis, cirrhosis, hepatic failure including cases resulting in transplantation or death (see WARNINGS & PRECAUTIONS) BEFORE INTERACTION.

INTERACTION

Alcohol and hepatotoxic medications may increase the risk hepatotoxicity. Antacids, anticholinergics/antispasmodics, histamine H2-antagonists or proton pump inhibitors will increase the pH of the stomach and decrease the absorption of ketoconazole. Use with astemizole, cisapride or terfenadine is contraindicated and may increase the risk of cardiac arrhythmias, including torsades de pointes. Didanosine contains a buffer to increase its absorption, which will decrease the absorption of ketoconazole. Concurrent use of rifampicin and isoniazid have been reported to decrease serum concentrations of ketoconazole. Ketoconazole has been reported to inhibit the metabolism of cyclosporin, which may increase the plasma concentration of cyclosporin to potentially toxic levels.
Concurrent use of ketoconazole and phenytoin may decrease the metabolism of phenytoin, resulting in increased plasma phenytoin concentrations and decreased plasma concentration of the antifungal which may lead to clinical failure or relapse of the fungal infection. Warfarin's anticoagulant effects may be increased in the presence of ketoconazole.
Ritonavir increases the bioavailability of ketoconazole. Therefore, when ritonavir is to be given concomitantly, higher doses (>200 mg/day) of ketoconazole tablets should not be used.
The use of ketoconazole (oral) may alter the following laboratory values: Alanine aminotransferase, alkaline phosphatase, aspartate aminotransferase, serum potassium, serum corticosteroid concentration including ACTH – induced levels, serum testosterone concentrations.
Effect of Ketoconazole on the metabolism of other drugs:
Ketoconazole can inhibit the metabolism of drugs metabolized by certain hepatic P-450 enzymes, especially the CPY3A family. This can result in an increase and/or prolongation of their effects including side effects.
Examples are:
Drugs that should not be used during treatment of ketoconazole: CYP3A4-metabolized HMG-CoA reductase inhibitors, eg. simvastatin and lovastatin. Drugs whose plasma levels, effects or side effects should be monitored. Their dosage if administered with ketoconazole should be reduced if necessary.
HIV protease inhibitors, eg idinavir, saquinavir
CPY3A4-metabolized calcium channel blockers, eg dihydropyridines.

PRESENTATION

Blister :10 x 10's.

STORAGE CONDITIONS AND SHELF LIFE

Store in a dry place below 30°C
Keep out of reach of children.
Jauhi daripada kanak-kanak.
Shelf life: Please refer to outer package.
Route Of Administration: Oral

Product Registration Holder:
Duopharma Manufacturing (Bangi) Sdn. Bhd.
Lot No. 2, 4, 6, 8 & 10, Jalan P/7, Section 13,
Bangi Industrial Estate, 43650 Bandar Baru Bangi,
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